

Week 2: functions, arrays, figures, calculus

Python syntax and code patterns. Keep it next to you while you work.

Writing a function

```
def sigmoid(x, slope=1.0, midpoint=0.0):
    """One line saying what it does."""
    return 1 / (1 + np.exp(-slope * (x - midpoint)))

sigmoid(0.5)           # defaults used
sigmoid(0.5, slope=4)  # say which one you mean
sigmoid(0.5, 4, 0.1)   # or give them in order

def two_things(x):
    return x.mean(), x.std(ddof=1)

m, s = two_things(values)  # unpack what comes back
```

Your own module

```
from src.qmn_utils import sigmoid, gaussian

# after editing the .py file, reload it:
import importlib, src.qmn_utils
importlib.reload(src.qmn_utils)
```

Making arrays

```
np.array([1.0, 2.0, 3.0])
np.linspace(0, 6, 600)      # 600 points, both ends included
np.arange(0, 6, 0.01)       # step 0.01, end excluded
np.zeros(10), np.ones(10)
np.full_like(t, 0.05)       # same shape, one value
x.shape, x.size, len(x)
dt = t[1] - t[0]            # the spacing of a grid
```

Maths on a whole array

```
y = 2 * x + 1               # every element at once
np.exp(x), np.log(x), np.sqrt(x)
np.sin(x), np.cos(x)
np.abs(x), np.pi, np.e
x.mean(), x.max(), x.min(), x.sum()
x.std(ddof=1)
np.allclose(a, b)           # equal to rounding error
```

Picking parts out

```
x[0], x[-1], x[10:20]
mask = x > 2                # array of True and False
x[mask]                     # only where True
x[(x > 0) & (x < 1)]        # brackets, & not "and"
np.argmax(y), np.argmin(y)  # position of the extreme
t[np.argmax(y)]             # where the peak happens
np.abs(x)[error <= 0.02].max()
```

The shapes of the week

```
y = amp * np.exp(-t / tau)      # decay
y = np.exp(-(x - mu)**2 / (2 * sigma**2)) # Gaussian
y = 1 / (1 + np.exp(-k * (x - x0))) # sigmoid
y = amp * np.sin(2 * np.pi * f * t) # sine
```

One figure

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(x, y, lw=2, ls="--", color="#d62728",
        label="tau = 1.5 s")
ax.axhline(0, color="gray", lw=0.8)
ax.axvline(x0, color="gray", lw=0.8)
ax.set_xlabel("time (s)")
ax.set_ylabel("amplitude")
ax.set_title("what this panel shows")
ax.set_xlim(0, 2); ax.set_ylim(-1, 1)
ax.legend(frameon=False, loc="lower left")
fig.tight_layout()
fig.savefig("figure.png", dpi=150, bbox_inches="tight")
plt.show()
```

Other kinds of panel

ax.scatter(x, y)	one dot per observation
ax.hist(v, bins=20, alpha=0.7)	a distribution
ax.bar(names, heights)	a value per category
ax.errorbar(x, y, yerr=e)	value with uncertainty
ax.text(x, y, "label")	writing on the axes

Several panels

```
fig, axes = plt.subplots(2, 1, figsize=(9, 5),
                          sharex=True)
axes[0].plot(t, signal); axes[0].set_ylabel("signal")
axes[1].plot(t, rate); axes[1].set_xlabel("time (s)")
fig.suptitle("one title for both")
fig.tight_layout()
```

Pattern: one curve per parameter

```
fig, ax = plt.subplots()
for tau in [0.5, 1.5]:
    ax.plot(t, exp_decay(t, tau=tau), lw=2,
            label=f"tau = {tau} s")
ax.legend(frameon=False)
```

Checking yourself

```
assert abs(a - b) < 1e-12, f"differ at x={value}"
np.allclose(a, b)           # True if equal to rounding

print(f"tau=0.5 gives {f(0.5):.3f}, "
      f"tau=1.5 gives {f(1.5):.3f}") # one f-string, two lines
```

Noise

```
rng = np.random.default_rng(7) # 7 is the seed
noise = rng.normal(0, 0.15, size=t.size)
trace = clean + noise
i0 = np.argmin(np.abs(x))       # index of the value nearest zero
```

Finding a point, finding peaks

```
above = y >= 0.8              # array of True and False
if above.any():               # ask first: does it ever happen?
    first = np.argmax(above)   # position of the first True
    x[first]                   # ... and where that is
# on an all False array argmax returns 0, silently

def smooth(y, w):             # moving average over w points
    pad = w // 2               # repeat the ends, never pad with
    zeros
    p = np.concatenate([np.full(pad, y[0]), y, np.full(pad,
y[-1])])
    return np.convolve(p, np.ones(w) / w, mode="valid")

from scipy.signal import find_peaks
rate = np.gradient(smooth(y, 5), x) # smooth, then
differentiate
peaks, properties = find_peaks(rate, height=0.005, distance=8)
x[peaks], rate[peaks]          # where the peaks are, how tall
```

Calculus, numerically

```
dy = np.gradient(y, x)        # slope at every point
d2y = np.gradient(dy, x)      # curvature

area = np.trapezoid(y, x)      # np.trapz on older NumPy
running = np.cumsum(y) * dt    # the accumulation
np.diff(y)                     # differences between neighbours
```

Pattern: repeat an estimate

```
rng = np.random.default_rng(7)

def estimate_many(noise_sd, n_repeats=100):
    estimates = []
    for _ in range(n_repeats):
        t, trace = make_transient(noise_sd, rng)
        estimates.append(recover_tau(t, trace))
    return np.array(estimates)

values = estimate_many(0.10)
print(f"mean {values.mean():.3f}, "
      f"spread {values.std(ddof=1):.3f}")
```